

WOLLONGONG, Australia

WMO: 957480

Lat: 34.5639S

Lon: 150.7900E

Elev: 9

StdP: 101.22

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.7	4.0	-1.2	3.4	13.9	-0.2	3.7	13.0	15.0	13.1	13.5	13.9	1.1	260	0.445

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.6	31.3	20.2	28.7	20.7	27.2	20.9	23.6	27.6	22.9	26.6	22.2	25.7	6.7	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	17.2	25.6	21.7	16.4	25.0	21.0	15.7	24.5	70.8	27.5	67.9	26.7	65.4	25.7	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.1	10.9	9.3	DB	0.2	38.0	1.0	2.8	-0.5	40.1	-1.1	41.7	-1.6	43.3	-2.3	45.3	
			WB	-0.2	24.9	1.0	0.8	-0.9	25.4	-1.4	25.9	-1.9	26.3	-2.6	26.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	21.6	21.3	20.1	17.5	14.6	12.6	11.8	12.4	14.4	16.4	18.3	19.9	(6)
	DBStd	4.09	2.47	2.17	2.06	2.11	1.95	1.86	2.01	2.06	2.51	2.70	2.59	2.48	(7)
	HDD10.0	9	0	0	0	0	0	2	5	2	0	0	0	0	(8)
	HDD18.3	962	2	1	7	39	116	172	203	184	122	73	32	11	(9)
	CDD10.0	2459	361	317	312	225	143	80	61	77	131	198	248	306	(10)
	CDD18.3	371	104	84	61	14	1	0	0	1	3	14	30	59	(11)
	CDH23.3	2343	666	454	279	89	12	0	2	5	69	161	227	379	(12)
	CDH26.7	534	158	82	39	14	0	0	0	0	16	53	72	99	(13)
(14) Wind	WSAvg	3.8	3.8	3.6	3.4	3.2	3.7	3.9	4.2	4.4	4.1	4.0	3.9	3.9	(14)
(15) Precipitation	PrecAvg	1125	101	121	122	109	90	101	45	79	61	81	114	80	(15)
	PrecMax	1876	464	488	477	446	394	747	194	657	262	314	775	291	(16)
	PrecMin	480	21	1	1	3	0	2	0	0	1	2	15	4	(17)
	PrecStd	356	81	113	100	114	88	128	45	124	63	77	127	76	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	33.2	30.6	28.7	24.8	21.3	22.3	23.8	28.8	32.3	33.2	33.6	(19)
		MCWB	23.1	23.9	21.8	17.4	15.9	13.9	13.4	13.3	15.9	17.3	19.3	18.7	(20)
	2%	DB	29.9	28.5	27.2	25.4	22.7	19.3	19.7	21.2	25.2	27.5	28.2	29.2	(21)
		MCWB	22.2	22.4	21.5	18.7	15.9	13.8	12.3	13.1	14.7	17.1	19.3	19.7	(22)
	5%	DB	27.9	27.0	25.9	24.0	21.1	18.3	18.1	19.3	22.4	24.5	25.5	26.7	(23)
		MCWB	22.6	22.0	21.0	18.4	15.3	13.2	11.6	12.5	14.4	17.0	19.8	20.6	(24)
	10%	DB	26.5	25.9	24.8	22.7	19.9	17.1	16.8	17.9	20.3	22.5	23.7	25.1	(25)
		MCWB	21.8	21.4	20.4	17.9	14.6	12.6	11.3	12.0	14.3	17.0	18.9	20.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.6	24.8	23.6	21.5	18.6	16.2	15.8	16.0	17.9	20.6	22.5	23.3	(27)
		MCDB	29.4	30.3	26.4	25.3	21.4	18.5	19.0	19.5	23.4	24.4	27.4	27.5	(28)
	2%	WB	23.8	23.6	22.6	20.2	17.3	15.0	14.0	14.7	16.9	19.5	21.4	22.3	(29)
		MCDB	28.0	27.2	25.8	23.4	20.5	17.4	16.9	18.5	21.4	23.6	25.6	26.2	(30)
	5%	WB	23.0	22.7	21.8	19.4	16.2	14.3	13.0	13.8	16.1	18.6	20.4	21.5	(31)
		MCDB	27.0	25.9	25.0	22.6	19.9	16.8	16.2	17.7	20.4	22.8	24.2	25.2	(32)
	10%	WB	22.3	22.0	21.0	18.5	15.3	13.6	12.0	12.8	15.0	17.6	19.5	20.7	(33)
		MCDB	25.9	24.9	24.2	21.6	19.0	16.1	15.7	16.6	19.2	21.7	22.9	24.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	8.0	8.5	10.0	10.6	9.4	10.6	11.1	11.4	10.6	9.4	9.1	(35)
		MCDBR	11.8	10.5	10.8	12.6	13.0	11.7	13.0	14.4	16.2	15.8	14.0	13.3	(36)
	5% WB	MCWBR	5.8	5.4	6.2	6.9	7.3	6.8	7.1	7.8	7.9	7.9	6.9	6.1	(37)
		MCWBR	10.0	9.1	9.6	11.0	12.0	9.5	11.2	12.5	13.3	12.4	11.3	10.6	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.400	0.389	0.365	0.336	0.318	0.308	0.298	0.307	0.339	0.357	0.388	0.402	(40)
		taud	2.430	2.487	2.528	2.584	2.584	2.610	2.622	2.589	2.500	2.482	2.430	2.400	(41)
	Ebn at Noon	Ebn at Noon	937	926	912	885	848	832	864	904	923	947	943	939	(42)
		Edh at Noon	122	111	100	84	74	67	70	81	100	111	121	127	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.48	5.49	4.69	3.79	2.98	2.32	2.73	3.62	4.87	5.72	6.08	6.63	(44)
	RadStd	RadStd	0.49	0.49	0.39	0.32	0.18	0.21	0.22	0.20	0.29	0.47	0.53	0.53	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (33 neighbors)	+0.43	N/A	N/A	+0.75	N/A	N/A	-1	-85	+182	+93

Nomenclature: See separate page